

Is Fusion Energy a Realistic Option for Morocco's Energy Mix by 2040?

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Abstract — Morocco imports 94% of its primary energy and remains heavily dependent on fossil fuels, creating both economic vulnerability and conflict with its climate commitments. While Morocco has set ambitious renewable energy targets, solar and wind remain intermittent, raising the question of whether fusion energy could offer a viable complement by 2040. This research examines the physical, technical, and economic feasibility of fusion energy for Morocco's energy mix, drawing on current fusion research timelines, cost projections, and comparisons with Morocco's existing energy infrastructure. The physics and engineering challenges of deuterium-tritium fusion are reviewed, along with the current state of major fusion programs including ITER and private initiatives such as Commonwealth Fusion Systems. A hypothetical 500 MW fusion plant is used to assess Morocco's grid capacity, water availability, land requirements, and financial feasibility, with direct comparisons to Morocco's existing solar, wind, and gas infrastructure, including the Noor Ouarzazate solar complex. The results show that fusion remains substantially more expensive than Morocco's current energy sources on a capital cost basis, and that no major fusion facility is expected to reach commercial operation before the late 2030s at the earliest. While fusion could theoretically provide dispatchable power that intermittent renewables cannot, Morocco's fiscal constraints, limited technical infrastructure, and existing renewable energy trajectory make large-scale fusion deployment unlikely by 2040. The research concludes that Morocco's most realistic path forward involves early-stage engagement, such as research partnerships and workforce development, rather than direct investment in fusion power generation.

fusion energy, Morocco, renewable energy, energy security, levelized cost of energy, energy transition, 2040

I. INTRODUCTION

Morocco follows the global trend of heading towards clean energy and investing in it as much as possible, having set the goal of reaching 52% of the energy mix from three priority resources (solar, wind, and hydraulic) by 2030. [1]

Morocco keeps working towards this goal, renewables accounted for only 9% of total energy supply in 2022, and the country imported 94% of its primary energy. This leaves Morocco is highly vulnerable to global price fluctuations and the political decisions of supplier nations. The energy supply is dominated by fossil fuels (58% oil, 32% coal and other solid fuels, and 1% gas), which generate carbon emissions and conflict with Morocco's international environmental commitments (e.g., the Paris Agreement). [2]

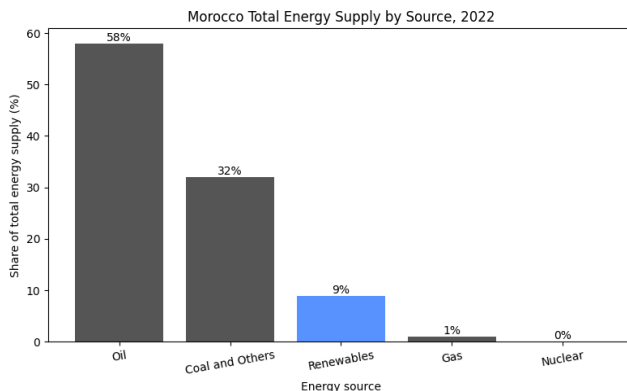


Fig. 1. Morocco Total Energy Supply by Source, 2022

Morocco is economically and politically sensitive, and its environment is under strain. These factors create the need for a new energy source. But is clean and renewable energy really enough to meet Morocco's needs? And should it still be Morocco's goal for 2040?

Renewable energy faces too many limitations, because solar and wind are not always available. Backup power (often from fossil fuels) is needed to fill the gaps, which still produces carbon emissions. Nuclear energy is a strong alternative, but it is too expensive to build: construction alone can take 10 to 15 years, and it requires massive upfront investment and strict safety and waste-management measures, all of which drive its cost far above solar or wind. This is where fusion energy comes in.

The light we see and the warmth we feel from every star in the universe come from a fusion reaction that happens right in the center of each star, like our star, the Sun. [3]

We all know the rule that the higher the temperature gets, the faster the atoms move. Well, in the core of our Sun, where the temperature reaches 15,000,000°C, hydrogen atoms collide and fuse at a speed high enough to overcome the natural electrostatic repulsion that exists between the positive charges of their nuclei. [3]

This fusion reaction produces a heavier element called helium and releases a really large amount of energy. Our Sun fuses about 600,000,000 tonnes of hydrogen into helium every second, with the help of its gravitational force. So to achieve this on Earth, we need to follow a different approach.

Fusion research in the 20th century proved that the most efficient fusion reaction in the laboratory is the reaction between two hydrogen isotopes, deuterium and tritium (D-T). D-T fusion produces the highest energy output at the lowest required temperature among fusion reactions. It requires a temperature of 150,000,000°C, which is 10 times hotter than the hydrogen reaction occurring in the Sun. [3]

Estimates of when fusion could become commercially viable vary widely. ITER, the largest international experiment, does not plan to begin deuterium-tritium operation until 2039 [6], while private companies such as Commonwealth Fusion Systems target a first commercial plant in the early 2030s [7].

Economic models such as the one used in the IAEA World Fusion Outlook 2025 assume commercial availability as early as 2035 [12]. This spread of dates, from optimistic industry targets to more conservative experimental timelines, is itself a key uncertainty for Morocco's 2040 goals.

The central question is therefore: Is Fusion a realistic option for Morocco's energy mix by 2040?

II. FUSION PHYSICS AND CURRENT STATUS

As global energy demand continues to grow, fusion has attracted attention because it is the energy of the stars: like them, it can produce enormous amounts of energy from a very small amount of fuel, giving it a virtually inexhaustible fuel supply. However, building such a machine on Earth is extremely difficult.

Fusion is a process in which two light atomic nuclei combine at a sufficiently high temperature and pressure. These two nuclei are isotopes of hydrogen: deuterium, which has one proton and one neutron, and tritium, which has one proton and two neutrons [3].

There is about one deuterium atom for every 5,000 hydrogen atoms in water, and this supply is enough to power humanity for the next billion years. Tritium, on the other hand, is much harder to obtain. It can be produced in certain types of nuclear reactors, or inside the fusion reactor itself by bombarding lithium with neutrons (a breeding blanket) [3].

When deuterium and tritium combine, they release energy because the products have slightly less mass than the reactants ($E = mc^2$), creating helium and a neutron [3].

The neutron carries most of the energy from this reaction (about 14.1 of the 17.6 MeV released). Unlike the helium nucleus (alpha particle), which is electrically charged and can be confined by magnetic fields, the neutron has no charge, so magnetic fields cannot contain it [3].

Several confinement approaches are being pursued to achieve fusion on Earth:

Magnetic confinement holds the superheated plasma inside strong magnetic fields, either in a doughnut-shaped tokamak or in a stellarator, whose magnetic field follows a twisted shape [4].

Inertial confinement compresses and heats a tiny fuel pellet with intense lasers, particle beams, or pulsed currents until it implodes and ignites [4].

Magneto-inertial fusion combines elements of both approaches [4].

These approaches face significant technological challenges, including:

Neutron damage and materials: the neutrons strike the reactor walls with high energy, causing physical damage over time, shortening the equipment's lifespan, and making the surrounding materials radioactive. This makes the safety and maintenance of the machine much more expensive [4].

Heat management and component resilience: removing the enormous heat from the plasma is a challenge in itself, and researchers are developing advanced cooling systems, plasma-facing materials that withstand extreme conditions, and more durable components [4].

Tritium cycle management: tritium, one of the two fuels, is scarce, so it must be produced, recycled, and retained inside the plant. Optimizing this cycle is essential for fuel supply, operational safety, and scaling up fusion systems [4].

Despite these challenges, significant progress has been made. In December 2022, the U.S. National Ignition Facility (NIF) achieved the first laboratory fusion experiment to release more energy than the laser energy delivered to the target, and in April 2025 it reached a record yield of 8.6 MJ from 2.08 MJ of laser energy [5]. However, this gain is measured on the target, not for the whole facility, and no device has yet delivered net electricity to a grid. ITER, the large international tokamak, plans to start deuterium-tritium operation in 2039 [6], while private companies such as Commonwealth Fusion Systems aim to connect a 400 MW plant to the grid in the early 2030s [7]. These are targets, not demonstrated results.

Governments and private companies in Europe, Russia, China, and the United States have also been investing billions of dollars in fusion [8]. By 2025, nearly 40 countries had dedicated fusion programmes, and more than 160 fusion devices were operating, under construction, or planned [9].

While fusion is included in global cost and deployment models, Africa appears there only as an aggregate demand region [12], and the only Middle Eastern involvement identified in this review is a financial investment in a foreign company [9], not a domestic feasibility assessment. No study was found that examines fusion energy specifically for an African or Middle Eastern country, including Morocco. This research addresses that gap.

No published study provides a direct LCOE (\$/MWh) figure for fusion; this comparison is therefore based on overnight capital cost (Section IV), which does not fully capture operating and fuel costs.

III. MOROCCO'S GRID AND FUSION COST OUTLOOK

For this analysis, we assume a hypothetical fusion power plant with a net electrical output of 500 MW, in line with the 300–500 MW of electrical output targeted by European DEMO (Demonstration Power Plant) design studies for the first generation of fusion power plants [10].

In 2024, ONEE reported that Morocco's total installed generation capacity reached 12,017 MW, a 4.7% increase compared to 11,474 MW in 2023. That same year, peak demand rose by 2.4% to 7,580 MW. This leaves a nominal reserve margin of approximately 4,437 MW (installed capacity minus peak demand) [11], about 8.9 times the assumed plant's output. However, this margin includes solar and wind, which had capacity utilisation of only 9% and 41% in 2023 [2]. Counting only non-renewable and hydro capacity (about 8,700 MW [2]), the firm margin falls to roughly 1,100 MW, so a 500 MW plant would use almost half of it. Since demand will keep growing toward 2040, these figures are only a present-day indicator and should be updated with projected 2040 demand.

A fusion reactor of this scale would require an estimated 15 hectares of land, based on a land requirement of roughly 300 m²/MW for fusion plants [14], though large-scale international facilities such as ITER occupy considerably more space (42 hectares) once laboratories and supporting infrastructure are included [15]. This is still far less land than solar or wind farms need to produce a comparable amount of power. Fusion would also require closed-loop cooling water, estimated at 1,514–2,725 L/MWh based on nuclear plant benchmarks, since no fusion-specific figures exist yet [16].

Morocco, however, already faces significant water stress. It is ranked among the countries most at risk of water shortage worldwide, and this is expected to worsen by 2040 [17]. Morocco's experience with large-scale energy infrastructure illustrates this concern. Noor I, part of the Noor Ouarzazate solar complex, uses a wet cooling system, while Noor II and III use dry cooling to reduce water use. Water for the complex is sourced from the El Mansour Eddahbi dam, about 12 km from the site, and stored in reservoirs with a capacity of 300,000 m³ [18].

For fusion to be viable in Morocco, siting and cooling choices would need to learn from this experience: sites should avoid competing directly with agricultural water use, and cooling technologies that minimize water consumption, similar to Noor's later phases, would likely be necessary.

Morocco does not yet have a dedicated fusion research programme, but there are early signs of capacity building in adjacent fields. UM6P's Physics School hosts plasma physics research through its ACER Center of Excellence [19], and in 2025 signed a partnership with the French nuclear technology company NAAREA to develop joint education and research programmes in reactor technology, including exchange programmes for students and doctoral candidates [20]. UM6P also has an active exchange partnership with TU Berlin's plasma and quantum research groups [21].

These partnerships suggest Morocco could build fusion-relevant expertise through international collaboration, similar to how smaller countries participate in ITER, but no domestic fusion-specific training programme currently exists.

IV. COMPARING THE ECONOMICS OF FUSION IN MOROCCO: IS FUSION WORTH IT?

The MIT Energy Initiative study, whose modelling was included in the IAEA World Fusion Outlook 2025 [9], does not predict when fusion will be commercially deployed or what it will cost. Instead, it explores the conditions under which fusion could be viable. In its base case, it assumes that fusion is commercially available in 2035 at an overnight capital cost of about \$11,000/kW (in 2021 US dollars), falling to about \$8,000/kW by 2050 and about \$4,300/kW by 2100 [12]. This assumption is more optimistic than ITER's schedule, since ITER, an experiment rather than a power plant, will not begin deuterium–tritium operation until 2039 [6]. The same model assumes that fusion plants start with a capacity factor of 40% in 2035, meaning that they produce only 40% of the electricity they would produce if they ran at full power all year, and that this rises gradually to 85% by 2060 [12]. Meanwhile, electricity consumption in Africa is projected to grow 9.3-fold between 2020 and 2100 [12].

However, the model is sensitive to cost. When fusion is expensive, it is deployed first in regions where it is comparatively cost-competitive, such as the United States and Europe, while uptake remains limited in other regions [12].

Fusion's \$11,000/kW cost in 2035 is 8 to 16 times higher than the \$700–1,400/kW typical of renewable energy, even though its capacity factor (40%, rising to 85% by 2060) is more than four times higher than Morocco's current solar capacity factor of 9% [2]. However, capacity factor alone does not determine the real cost of electricity produced: using the overnight capital cost figures above, \$11,000 invested in solar at Morocco's 2023 performance would generate approximately 8,672 kWh per year, compared to approximately 3,504 kWh per year for the same investment in fusion at its 2035 capacity factor [12]. In other words, despite its much lower capacity factor, solar remains roughly 2.5 times cheaper per unit of electricity produced on this simplified basis, because its capital cost is so much lower. This is a simplified comparison based only on overnight capital costs; it does not include operating costs, fuel, or plant lifetime, and it uses Morocco's current (2023) solar capacity factor rather than a projected 2035 figure, which is not available for Morocco. If Morocco's solar performance improves by 2035, the cost gap would narrow further.

For Morocco, this suggests that fusion is unlikely to be an early adopter option. Renewables, mainly wind, solar, and hydro, already make up 39% of Morocco's installed electricity capacity but only 19% of its electricity generation, and the country imports 94% of its primary energy [2]. If fusion remains expensive, Morocco would probably use it

only in a limited way, if at all. Fusion would become relevant for Morocco mainly if its cost falls, and this is the key uncertainty for the 2040 question.

No published study provides a direct LCOE (levelized cost of energy) figure for fusion; the only available projections, from MIT and the IAEA, report overnight capital costs only. By comparison, a recent study of Morocco's existing electricity mix found that gas-fired plants have an LCOE of about \$50-90/MWh, the lowest among dispatchable sources, while solar LCOE in Morocco ranges from roughly \$200-730/MWh depending on the technology and design used [22]. Without a comparable LCOE figure for fusion, this section relies on the capital cost comparison above rather than a full levelized cost comparison. This reflects a limitation in the available data rather than a limitation of the analysis itself.

Fusion investment is not limited to governments. Private companies and sovereign-backed funds are also involved. For example, Zap Energy received a strategic investment from Plynth Energy, a fund backed by the United Arab Emirates, which is one of the first direct sovereign investments from the Middle East in fusion [9]. Beyond this, other financing models exist that could be relevant for Morocco. In large international projects such as ITER, member countries contribute mostly "in-kind" rather than in cash: about 90% of each member's contribution is delivered as completed components, systems, or expertise rather than money, which allows smaller or less wealthy countries to participate through industrial capacity rather than large cash budgets [23]. Italy's national fusion programme illustrates a mixed financing model, combining domestic government funding (about 80%), contributions from the European fusion consortium EUROfusion, and a loan from the European Investment Bank [24]. Globally, private investment in fusion companies has also grown substantially, reaching over \$4.7 billion by 2022 [24].

Given Morocco's current fiscal position, a large direct capital investment in fusion technology is unlikely to be realistic before 2040. However, the in-kind and mixed-financing models used by ITER and Italy suggest that participation through research infrastructure, workforce training, or hosting agreements, rather than full project funding, may be a more realistic path for Morocco to engage with fusion development. This capital cost comparison does not, however, capture fusion's main advantage: unlike solar and wind, which are intermittent, fusion could provide firm, dispatchable power available around the clock, similar to how fission plants operate today — a capability Morocco's current renewable mix cannot offer on its own.

V. FUSION VS. SOLAR: WHEN DOES FUSION MAKE SENSE FOR MOROCCO?

This comparison does not, however, capture fusion's main advantage: unlike solar and wind, which are intermittent, fusion could provide firm, dispatchable power available around the clock, similar to how fission plants

operate today. This is a capability Morocco's current renewable mix cannot offer on its own.

Morocco already aims to meet up to 90% of its electricity needs through renewable energy by 2035, largely through accelerated deployment of decentralized solar systems [13], the same year fusion is assumed to become commercially available in optimistic models. This raises the question, addressed further in the Discussion, of whether investing in fusion would be a better use of resources than continuing to expand the renewable systems Morocco is already deploying successfully.

Noor demonstrates that not all solar technologies are equally cost-effective; large-scale CSP with storage, while providing more dispatchable power than simple PV, comes at a capital cost comparable to or even higher than fusion's 2035 projections. This suggests fusion's disadvantage may be less about cost alone and more about technological readiness and deployment timeline (Sections II-III) [25].

Directly comparing fusion to Morocco's flagship solar project offers a more concrete picture than a general solar comparison. The Noor Ouarzazate complex (Noor I-III) cost an estimated \$9 billion to build for 510 MW of capacity, or roughly \$17,647/kW [25], higher than fusion's projected \$11,000/kW in 2035 [12]. Noor's combined annual output of 1,470 GWh implies a capacity factor of about 33% [25], well above the 9% average for Moroccan solar overall [2], since Noor uses concentrated solar power (CSP) with thermal storage rather than simple photovoltaic panels. Using the same \$17,647 investment, Noor would generate approximately 2,882 kWh/year at its actual capacity factor, compared to approximately 5,606 kWh/year for fusion at its 2035 capacity factor of 40%, meaning fusion could, in principle, outperform Noor's specific technology per dollar invested, unlike the earlier comparison with generic solar PV in Section IV. This suggests fusion's main disadvantage relative to Morocco's most advanced solar project may lie less in cost than in technological readiness and deployment timeline (Sections II-III).

VI. BARRIERS AND RISKS: DEPENDENCY AND UNCERTAINTY

If Morocco were to depend entirely on a single foreign country or company for fusion technology, this would create a risk similar to the current dependency on imported fossil fuels. One way to reduce this risk is for Morocco to position itself as a strategic partner rather than a technology buyer, for example by hosting testing or research facilities in exchange for technology transfer and training, similar to how smaller countries have joined international programmes such as ITER [9].

VII. CONCLUSION

Bringing together the evidence from this research, fusion energy is unlikely to be a realistic option for Morocco's electricity mix by 2040, though the picture is more nuanced than a simple yes or no.

From a technological standpoint, fusion is not expected to be commercially deployable by 2040 under any but the most optimistic scenarios. ITER, the world's largest fusion experiment, will not even begin deuterium-tritium operation until 2039, and private companies target only early demonstration plants in the early 2030s, not commercial deployment at scale (Section II). Economic models that assume 2035 commercial availability, such as the one used by the IAEA and MIT, are themselves more optimistic than ITER's own schedule (Section IV).

Economically, fusion's overnight capital cost in 2035 (\$11,000/kW) remains far higher than Morocco's existing gas and coal plants, and even higher than Morocco's flagship Noor solar complex when capacity factor is taken into account, fusion could in principle be more efficient per dollar invested than Noor's specific technology, but this advantage only holds under optimistic 2035 cost assumptions that have not yet been demonstrated (Sections III-V). Morocco's own fiscal position makes a large direct investment unlikely; more realistic financing paths would resemble the in-kind or mixed-financing models used by ITER and Italy, rather than full project ownership (Section IV).

Technically, Morocco's grid could theoretically absorb a fusion plant of this scale, though the country's firm generation margin is tighter than it first appears once intermittent renewables are excluded (Section III). Morocco also faces meaningful water stress, and any fusion plant would need to adopt dry-cooling approaches similar to those used in Noor's later phases to avoid competing with agricultural water use (Section III). Morocco currently has no dedicated fusion research programme, though early academic partnerships, such as UM6P's collaboration with NAAREA and TU Berlin, suggest the beginnings of relevant technical capacity (Section III).

Strategically, fusion could offer Morocco something its renewable-heavy grid currently cannot: firm, dispatchable power (Section V). But Morocco is already pursuing an ambitious and comparatively low-cost renewable pathway, aiming for up to 90% renewable electricity by 2035 (Section IV), and depending entirely on foreign fusion technology would recreate the same type of external dependency Morocco is currently trying to escape with fossil fuel imports (Section VI).

Taken together, fusion is not a realistic contributor to Morocco's energy mix by 2040. It is, at best, a technology Morocco could begin preparing for during this period, through research partnerships, workforce training, and monitoring of global cost trends, rather than one it could deploy.

Based on this analysis, several concrete steps could position Morocco to benefit from fusion energy if and when it becomes commercially viable, without requiring the country to make a premature large-scale investment.

First, Morocco could pursue an in-kind participation model similar to that used by ITER member states, contributing research capacity or hosting infrastructure rather than large cash sums (Section IV). UM6P's existing partnerships with NAAREA and TU Berlin offer a starting point for this kind of engagement, and could be expanded specifically toward plasma physics and fusion-relevant research (Section III).

Second, Morocco should continue to invest in the workforce and academic capacity needed to eventually operate or collaborate on fusion projects, building on UM6P's physics programmes and existing international exchange agreements (Section III).

Third, given the significant uncertainty in the cost projections discussed in this research (Section IV), Morocco should monitor global fusion cost trends over the coming decade rather than commit to a specific technology or partner prematurely. If fusion's capital costs fall substantially, as some models project, the case for Moroccan involvement would strengthen considerably by the 2040s (Section IV).

Finally, any future fusion project in Morocco should apply lessons already learned from Noor: siting decisions and cooling technologies should be chosen to avoid competing with agricultural water use, particularly given Morocco's existing water stress (Section III).

None of these steps require Morocco to commit to building a fusion reactor by 2040. Rather, they would allow Morocco to be positioned to act quickly once fusion technology, and its costs, become clearer.

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